

Dušan Korizma

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LINKS

[GitHub](#), [LinkedIn](#)

PROFILE

I am a passionate student who loves coding, algorithms, and problem-solving. My curiosity drives me to explore computer science and refine my skills. I'm fascinated by mathematics and its applications in optimization, combinatorics, and theory.

EDUCATION

Oct 2023 — Present	Software Engineering, School of Electrical Engineering Currently enrolled as a second-year student, with a grade average of 9.75.	Belgrade
Sep 2019 — Jun 2023	Mathematical Grammar School Graduated from the Mathematical Grammar School, excelling in mathematics, physics, and computer science through a rigorous, competitive curriculum.	Belgrade

EMPLOYMENT HISTORY

Oct 2023 — Present	Junior Assisting Teacher, Petnica Science Center Creating and organizing lectures, projects and workshops for high school students interested in Computer Science.
Nov 2024 — Present	Mentor, Petlja Online programming school for elementary and high school students, "Raspetljavanje". Organizing weekly meetings, tracking students' progress and providing consultancy space.

PROJECTS

Analysis of Genetic Algorithm Methods for Deriving a Function Formula from a Given Graph

Developed a Python-based system to extrapolate mathematical formulas from graphical data using binary trees and genetic algorithms. The project combined evolutionary optimization and tree-based representations to accurately derive relationships from complex datasets.

B* Tree Implementation

Implemented a B* tree index structure in C++ for an SQL-like database, enabling efficient indexing, searching, and data management. The project was developed as part of a university course on advanced data structures and algorithms.

NEAT (NeuroEvolution of Augmenting Topologies)

In my graduation paper, I studied the NEAT (NeuroEvolution of Augmenting Topologies) algorithm and applied it to evolve neural networks for optimizing gameplay in the Snake game. Using inputs like wall proximity, food position, and movement direction, the neural networks were trained to improve decision-making, with a custom fitness function rewarding survival and food consumption.

Command Line Interpreter

Implemented a command-line interpreter in C++ with support for input/output redirection, piping, and commands like echo, time, touch, rm, and wc. Enhanced proficiency in system calls, process management, and advanced C++ programming.

SKILLS

SQL

C/C++